THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 29	$(\alpha+\beta)^2-2\alpha\beta$
2	**(a)** 24	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
3	**(c)** 6	$D=0$ gives $k^2=4m$
4	**(b)** 2	Substitute the root
5	**(c)** $k\le 3$ and $k\neq 0$	$D=36-12k\ge 0$
6	**(b)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
7	**(d)** 53	$(\alpha+\beta)^2-2\alpha\beta$
8	**(d)** 32	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$
9	**(d)** 8	$D=0$ gives $k^2=4m$
10	**(a)** 2	Substitute the root

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$D=(4)^2-4k(1)=0$ (1); $k=4$ (1).	1+1
12	$\alpha+\beta=6$, $\alpha\beta=5$ (1); $\alpha^2+\beta^2=6^2-2(5)=26$ (1).	1+1
13	Let number x: $x+\frac{1}{x}=\frac{5}{2}$ (1); $x=2$ or $x=\frac{1}{2}$ (1).	1+1
14	$D=(6)^2-4k(3)=0$ (1); $k=3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $=\frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $3(x^2+1)=5(x^2-1)$ (1); $x^2=4$ so $x=\pm 2$ (1).	1+1+1
16	$D=25-4k\ge 0$ (1); $k\le\frac{25}{4}$ (1); greatest integer $k=6$ (1).	1+1+1
17	Integers x, $x+2$: $x^2+(x+2)^2=74$ (1); $x^2+2x-35=0$ (1); $x=5$; integers 5 and 7 (1).	1+1+1
	OR: LHS $=\frac{2(x^2+4)}{x^2-4}$ (1); cross-multiplying: $5(x^2+4)=13(x^2-4)$ (1); $x^2=9$ so $x=\pm 3$ (1).	1+1+1
18	$D=49-4k\ge 0$ (1); $k\le\frac{49}{4}$ (1); greatest integer $k=12$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2+(x+2)^2=130$ (1); $x^2+2x-63=0$ (1); $x=7$; integers 7 and 9 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Smaller tap x h, larger $x-5$ h: $\frac{1}{x}+\frac{1}{x-5}=\frac{1}{\frac{10}{3}}$ (2); simplify to a quadratic and solve, rejecting the root less than 5: $x=10$ (2); times 10 h and 5 h (1).	2+2+1
	OR: Boat speed x: $\frac{48}{x-2}+\frac{48}{x+2}=7$ (1); $48(2x)=7(x^2-4)$ (1); roots of the resulting quadratic: $x=14$ or negative (1); speed 14 km/h (1).	1+1+1+1
20	Smaller tap x h, larger $x-4$ h: $\frac{1}{x}+\frac{1}{x-4}=\frac{1}{\frac{24}{5}}$ (2); simplify to a quadratic and solve, rejecting the root less than 4: $x=12$ (2); times 12 h and 8 h (1).	2+2+1
	OR: Boat speed x: $\frac{60}{x-4}+\frac{60}{x+4}=8$ (1); $60(2x)=8(x^2-16)$ (1); roots of the resulting quadratic: $x=16$ or negative (1); speed 16 km/h (1).	1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $k \le 2$ and $k \neq 0$	$D = 16 - 8k \ge 0$
2	**(c)** $\frac{7}{12}$	$\frac{\alpha+\beta}{\alpha\beta}$
3	**(b)** 58	$(\alpha+\beta)^2 - 2\alpha\beta$
4	**(b)** 40	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
5	**(b)** 10	$D = 0$ gives $k^2 = 4m$
6	**(a)** -2	Substitute the root
7	**(b)** $k \le 4$ and $k \neq 0$	$D = 64 - 16k \ge 0$
8	**(c)** $\frac{9}{20}$	$\frac{\alpha+\beta}{\alpha\beta}$
9	**(c)** 73	$(\alpha+\beta)^2 - 2\alpha\beta$
10	**(d)** 63	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\alpha+\beta = 8$, $\alpha\beta = 7$ (1); $\alpha^2+\beta^2 = 8^2 - 2(7) = 50$ (1).	1+1
12	Let number x: $x + \frac{1}{x} = \frac{10}{3}$ (1); $x = 3$ or $x = \frac{1}{3}$ (1).	1+1
13	$D = (8)^2 - 4k(8) = 0$ (1); $k = 2$ (1).	1+1
14	$\alpha+\beta = 9$, $\alpha\beta = 8$ (1); $\alpha^2+\beta^2 = 9^2 - 2(8) = 65$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $8(x^2+1) = 10(x^2-1)$ (1); $x^2 = 9$ so $x = \pm 3$ (1).	1+1+1
16	$D = 81 - 4k \ge 0$ (1); $k \le \frac{81}{4}$ (1); greatest integer $k = 20$ (1).	1+1+1
17	Integers x, $x+2$: $x^2 + (x+2)^2 = 202$ (1); $x^2 + 2x - 99 = 0$ (1); $x = 9$; integers 9 and 11 (1).	1+1+1
	OR: LHS $= \frac{2(x^2+9)}{x^2-9}$ (1); cross-multiplying: $16(x^2+9) = 34(x^2-9)$ (1); $x^2 = 25$ so $x = \pm 5$ (1).	1+1+1
18	$D = 121 - 4k \ge 0$ (1); $k \le \frac{121}{4}$ (1); greatest integer $k = 30$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2 + (x+2)^2 = 290$ (1); $x^2 + 2x - 143 = 0$ (1); $x = 11$; integers 11 and 13 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Smaller tap x h, larger $x-5$ h: $\frac{1}{x} + \frac{1}{x-5} = \frac{1}{6}$ (2); simplify to a quadratic and solve, rejecting the root less than 5: $x = 15$ (2); times 15 h and 10 h (1).	2+2+1
	OR: Speed x: $\frac{840}{x} - \frac{840}{x+10} = 2$ (1); $8400 = 2x(x+10)$ so $x^2 + 10x - 4200 = 0$ (2); $(x-60)(x+70) = 0$ (1); speed 60 km/h (1).	1+2+1+1
20	Smaller tap x h, larger $x-3$ h: $\frac{1}{x} + \frac{1}{x-3} = \frac{1}{\frac{18}{5}}$ (2); simplify to a quadratic and solve, rejecting the root less than 3: $x = 9$ (2); times 9 h and 6 h (1).	2+2+1
	OR: Speed x: $\frac{1080}{x} - \frac{1080}{x+5} = 3$ (1); $5400 = 3x(x+5)$ so $x^2 + 5x - 1800 = 0$ (2); $(x-40)(x+45) = 0$ (1); speed 40 km/h (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	$D = 0$ gives $k^2 = 4m$
2	**(b)** 2	Substitute the root
3	**(c)** $k \leq 5$ and $k \neq 0$	$D = 100 - 20k \geq 0$
4	**(b)** $\frac{8}{15}$	$\frac{\alpha+\beta}{\alpha\beta}$
5	**(c)** 40	$(\alpha+\beta)^2 - 2\alpha\beta$
6	**(c)** 65	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
7	**(a)** 14	$D = 0$ gives $k^2 = 4m$
8	**(a)** 2	Substitute the root
9	**(a)** $k \leq 1$ and $k \neq 0$	$D = 4 - 4k \geq 0$
10	**(d)** $\frac{11}{18}$	$\frac{\alpha+\beta}{\alpha\beta}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let number x: $x + \frac{1}{x} = \frac{17}{4}$ (1); $x = 4$ or $x = \frac{1}{4}$ (1).	1+1
12	$D = (10)^2 - 4k(5) = 0$ (1); $k = 5$ (1).	1+1
13	$\alpha + \beta = 10$, $\alpha\beta = 9$ (1); $\alpha^2 + \beta^2 = 10^2 - 2(9) = 82$ (1).	1+1
14	Let number x: $x + \frac{1}{x} = \frac{26}{5}$ (1); $x = 5$ or $x = \frac{1}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LHS $= \frac{2(x^2+4)}{x^2-4}$ (1); cross-multiplying: $21(x^2+4) = 29(x^2-4)$ (1); $x^2 = 25$ so $x = \pm 5$ (1).	1+1+1
16	$D = 169 - 4k \geq 0$ (1); $k \leq \frac{169}{4}$ (1); greatest integer $k = 42$ (1).	1+1+1
17	Integers x, $x+2$: $x^2 + (x+2)^2 = 394$ (1); $x^2 + 2x - 195 = 0$ (1); $x = 13$; integers 13 and 15 (1).	1+1+1
	OR: LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $15(x^2+1) = 17(x^2-1)$ (1); $x^2 = 16$ so $x = \pm 4$ (1).	1+1+1
18	$D = 225 - 4k \geq 0$ (1); $k \leq \frac{225}{4}$ (1); greatest integer $k = 56$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2 + (x+2)^2 = 514$ (1); $x^2 + 2x - 255 = 0$ (1); $x = 15$; integers 15 and 17 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Boat speed x: $\frac{36}{x-3} + \frac{36}{x+3} = \frac{32}{5}$ (1); $36(2x) = \frac{32}{5}(x^2-9)$ (1); roots of the resulting quadratic: $x = 12$ or negative (1); speed 12 km/h (1).	1+1+1+1
	OR: Speed x: $\frac{360}{x} - \frac{360}{x+12} = 1$ (1); $4320 = x(x+12)$ so $x^2 + 12x - 4320 = 0$ (2); $(x-60)(x+72) = 0$ (1); speed 60 km/h (1).	1+2+1+1
20	Boat speed x: $\frac{40}{x-5} + \frac{40}{x+5} = 6$ (1); $40(2x) = 6(x^2-25)$ (1); roots of the resulting quadratic: $x = 15$ or negative (1); speed 15 km/h (1).	1+1+1+1
	OR: Speed x: $\frac{720}{x} - \frac{720}{x+4} = 2$ (1); $2880 = 2x(x+4)$ so $x^2 + 4x - 1440 = 0$ (2); $(x-36)(x+40) = 0$ (1); speed 36 km/h (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.